CIS 678 - Machine Learning

– Logistic Regression –

A Probabilistic Linear Classifier

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Last Class

- Classification vs. Regression
- Linear Regression
  - Linear Model
  - Least Squares
  - MLE and Regularization
- From Regression to Classification
- Perceptron

Agenda

- Generative vs. Discriminative Classifiers
- Why Logistic Regression?
- Sigmoid Function and Logistic Regression Model
- Linear Decision Boundary
- Maximum Likelihood Estimation
- Logistic Loss and Gradient Descent
- MAP and Regularization

Naive Bayes Recap

Classification Goal

For an input (x), predict the class with the largest posterior probability:

$$f^*(x) = \arg\max_y P(Y = y \mid X = x)$$

Naive Bayes Assumption

Given the class (Y), the features are conditionally independent:

$$P(X_1, \ldots, X_d \mid Y) = \prod_{i=1}^{d} P(X_i \mid Y)$$

Using Bayes' rule: $P(Y \mid X) \propto P(Y) \prod_{i=1}^{d} P(X_i \mid Y)$

Therefore:

$$f_{NB}(x) = \arg\max_y P(Y = y) \prod_{i=1}^{d} P(X_i = x_i \mid Y = y)$$



How Do We Learn the Model?

from the training data using MLE or MAP.

Generative vs. Discriminative Classifiers

Generative Classifiers

Example: **Naive Bayes**

Model the joint distribution: $P(X, Y) = P(X \mid Y)P(Y)$

- Learn $P(X \mid Y)$
- Learn $P(Y)$
- Use Bayes' rule to obtain $P(Y \mid X)$

$$P(Y \mid X) \propto P(X \mid Y)P(Y)$$

Discriminative Classifiers

Example: **Logistic Regression**

Model the class probability directly: $P(Y \mid X)$

or learn the decision boundary directly.

- Do not need to model how X is generated
- Focus directly on predicting Y from X

Why Logistic Regression?

Hard Class Prediction

A linear classifier such as the Perceptron predicts only a class: $\hat{y} \in \{0, 1\}$

- Example: Spam

But it does not tell us how confident the prediction is.

Probability Prediction

What if we want: $P(Y = 1 \mid X = x)$

- Example: $P(\text{Spam} \mid x) = 0.92$



How can we convert a linear score into a probability between 0 and 1?

This gives us both:

- a probability
- a class prediction after applying a threshold $P(Y = 1 \mid x) \geq 0.5 \quad \Rightarrow \quad \hat{y} = 1$

Logistic Regression: Sigmoid Function



How can we convert a linear score into a probability between 0 and 1?

The linear score $z = w^T x + b$ can take any real value: $-\infty < z < +\infty$

We need a function that maps this score to a value between 0 and 1

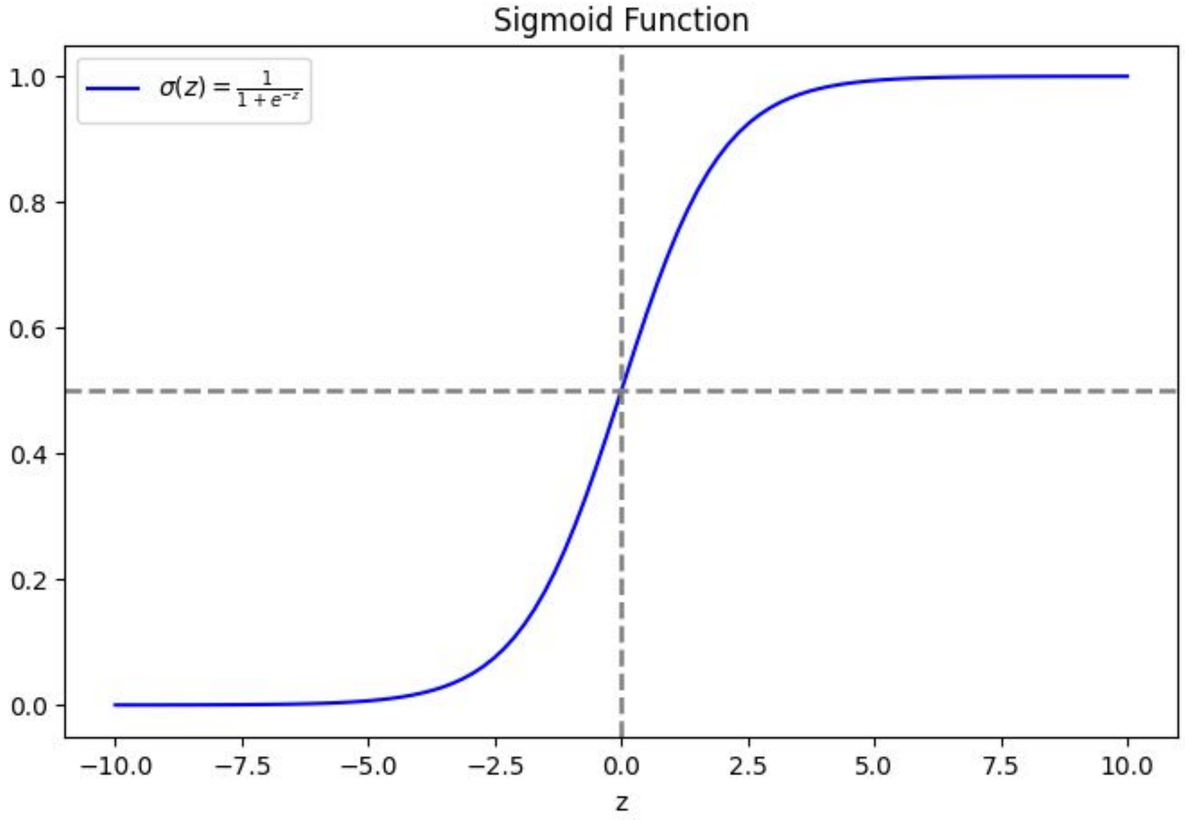
Sigmoid Function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\sigma(z) \in (0, 1)$$

The sigmoid converts a linear score into a probability-like value.

- $z = 0 \Rightarrow \sigma(z) = 0.5$
- $z \gg 0 \Rightarrow \sigma(z) \approx 1$
- $z \ll 0 \Rightarrow \sigma(z) \approx 0$



Logistic Regression Model

First $z = w^T x + b$



Since the sigmoid function is nonlinear, is Logistic Regression a linear classifier or a nonlinear classifier?

Then $P(Y = 1 \mid x) = \sigma(w^T x + b) = \dfrac{1}{1 + e^{-(w^T x + b)}}$

For binary classification: $P(Y = 0 \mid x) = 1 - P(Y = 1 \mid x)$

Class Prediction

Using a threshold of 0.5 :

$$\hat{y} = \begin{cases} 1, & P(Y = 1 \mid x) \geq 0.5 \\ 0, & P(Y = 1 \mid x) < 0.5 \end{cases}$$

Logistic Regression is a Linear Classifier!

Recall: $P(Y = 1 \mid x) = \sigma(w^T x + b) = \dfrac{1}{1 + e^{-(w^T x + b)}}$

Using threshold of 0.5 :

$$P(Y = 1 \mid x) \geq 0.5$$

Since $\sigma(0) = 0.5$

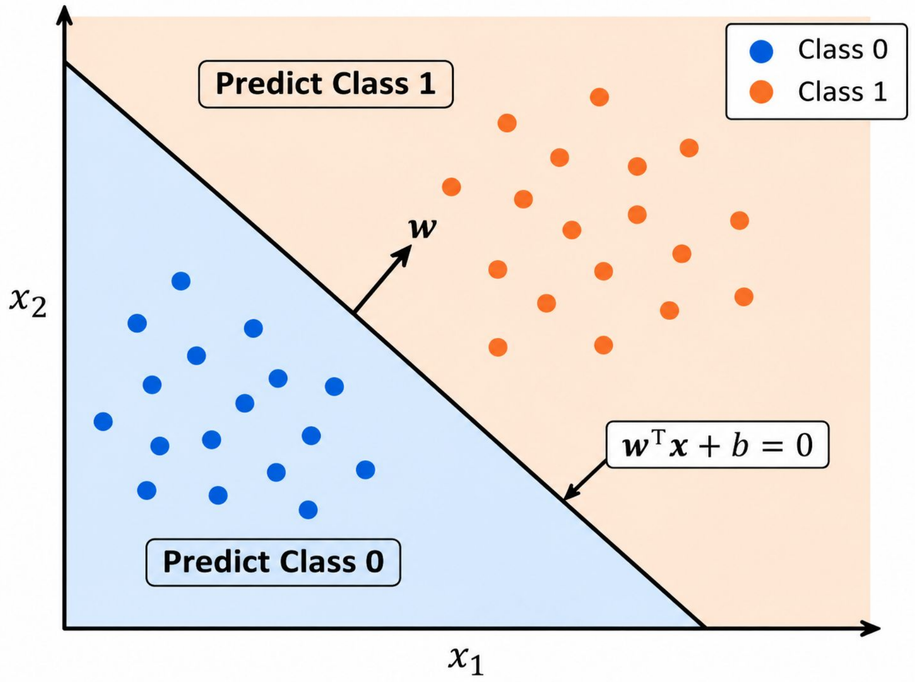
$$\Big\} \; P(Y = 1 \mid x) \geq 0.5 \iff w^T x + b \geq 0$$

So, the decision boundary is

$$w^T x + b = 0$$

- $w^T x + b > 0$: predict class 1
- $w^T x + b < 0$: predict class 0

Although the **sigmoid function is nonlinear**, the **decision boundary is linear**.



Logistic Regression: Practice

Given

$$w = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \qquad x = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \qquad b = 0.5$$

Using Logistic Regression,

$$P(Y = 1 \mid x) = \sigma(w^T x + b)$$

with classification threshold (0.5).



What is the predicted class $\hat{y} = ?$

Logistic Regression: Learning the Parameters



How do we learn w and b from the training data?

Suppose the training set is $\{(x_i, y_i)\}_{i=1}^{n}, \qquad y_i \in \{0, 1\}$

For each example, $p_i = P(Y = 1 \mid x_i) = \sigma(w^T x_i + b)$

Therefore, $P(y_i \mid x_i) = p_i^{y_i}(1 - p_i)^{1-y_i}$

For all training examples, the likelihood is

$$L(w, b) = \prod_{i=1}^{n} p_i^{y_i}(1 - p_i)^{1-y_i}$$

Maximum Likelihood Estimation

Choose w and b that make the observed training labels as likely as possible:

$$(w^*, b^*) = \arg\max_{w,b} L(w, b)$$

Log-Likelihood and Logistic Loss

$$L(w,b) = \prod_{i=1}^{n} p_i^{y_i}(1-p_i)^{1-y_i} \quad \text{where} \quad p_i = \sigma(w^T x_i + b)$$

Instead of maximizing the product, take the logarithm:

$$\log L(w,b) = \sum_{i=1}^{n} [y_i \log p_i + (1-y_i)\log(1-p_i)]$$

Maximum Likelihood:

$$\max_{w,b} \log L(w,b)$$

=

Minimize the negative log-likelihood:

$$\mathcal{L}(w,b) = -\sum_{i=1}^{n} [y_i \log p_i + (1-y_i)\log(1-p_i)]$$

binary cross-entropy loss

Gradient Descent for Logistic Regression

Objective (Logistic Loss)

$$\mathcal{L}(\mathbf{w}, b) = -\sum_{i=1}^{n} \left[y_i \log p_i + (1 - y_i) \log (1 - p_i) \right]$$

$$p_i = \sigma(\mathbf{w}^T \mathbf{x}_i + b)$$

Gradients

$$\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = \sum_{i=1}^{n} (p_i - y_i) \mathbf{x}_i \qquad \frac{\partial \mathcal{L}}{\partial b} = \sum_{i=1}^{n} (p_i - y_i)$$

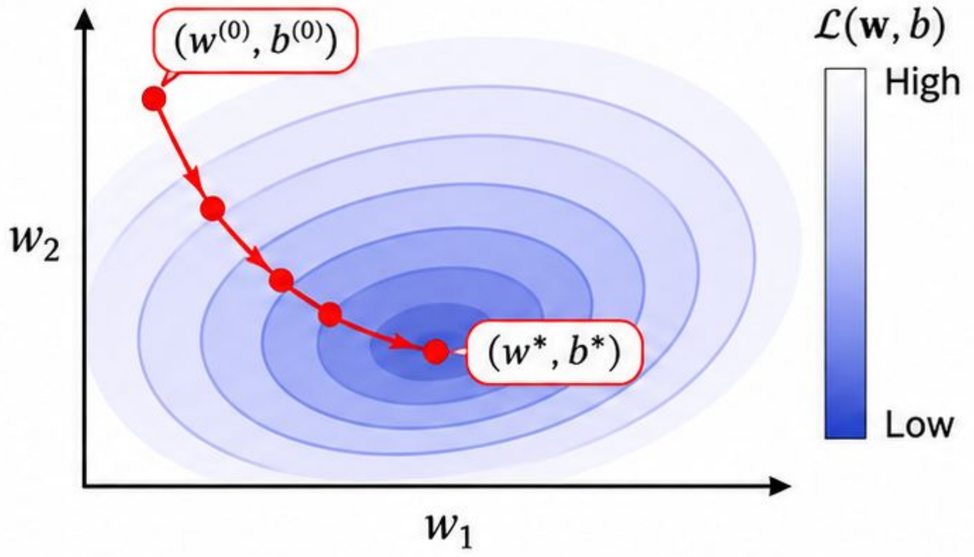
Gradient Descent Update

$$\mathbf{w} \leftarrow \mathbf{w} - \eta \frac{\partial \mathcal{L}}{\partial \mathbf{w}} = \mathbf{w} - \eta \sum_{i=1}^{n} (p_i - y_i) \mathbf{x}_i$$

$$b \leftarrow b - \eta \frac{\partial \mathcal{L}}{\partial b} = b - \eta \sum_{i=1}^{n} (p_i - y_i)$$

η: learning rate (a small positive number)

Gradient Descent (Illustration)



Algorithm (Batch Gradient Descent)

1. Initialize $\mathbf{w}, b$ (e.g., $\mathbf{w} = 0, b = 0$).
2. Repeat until convergence:
   - Compute the gradients $\frac{\partial \mathcal{L}}{\partial \mathbf{w}}, \frac{\partial \mathcal{L}}{\partial b}$.
   - Update $\mathbf{w}$ and b using the rules above.

MAP → Regularization

Maximum Likelihood

MLE chooses parameters that maximize the likelihood:

$$(w^*, b^*)_{\text{MLE}} = \arg\max_{w,b} P(D \mid w, b)$$

Maximum A Posteriori Estimation

MAP also includes a prior belief about the parameters:

$$P(w \mid D) \propto P(D \mid w)P(w)$$



$$w^*_{\text{MAP}} = \arg\max_{w} \left[\log P(D \mid w) + \log P(w)\right]$$

Gaussian Prior $w \sim \mathcal{N}(0, \tau^2 I)$

Then MAP estimation is equivalent to minimizing $\text{Logistic Loss} + \lambda \|w\|_2^2$

Summary

- Logistic Regression is a discriminative probabilistic classifier.
- Model: $z = w^T x + b$, then $P(Y = 1 \mid x) = \sigma(z)$
- Sigmoid converts the score into a probability:
- Prediction: threshold the probability to obtain a class
- Decision boundary: $w^T x + b = 0$, so Logistic Regression is a linear classifier
- Learning: MLE ⇐⇒ minimizing logistic loss
- Optimization: Gradient Descent
- Regularization: MAP + Gaussian prior ⇒ regularization